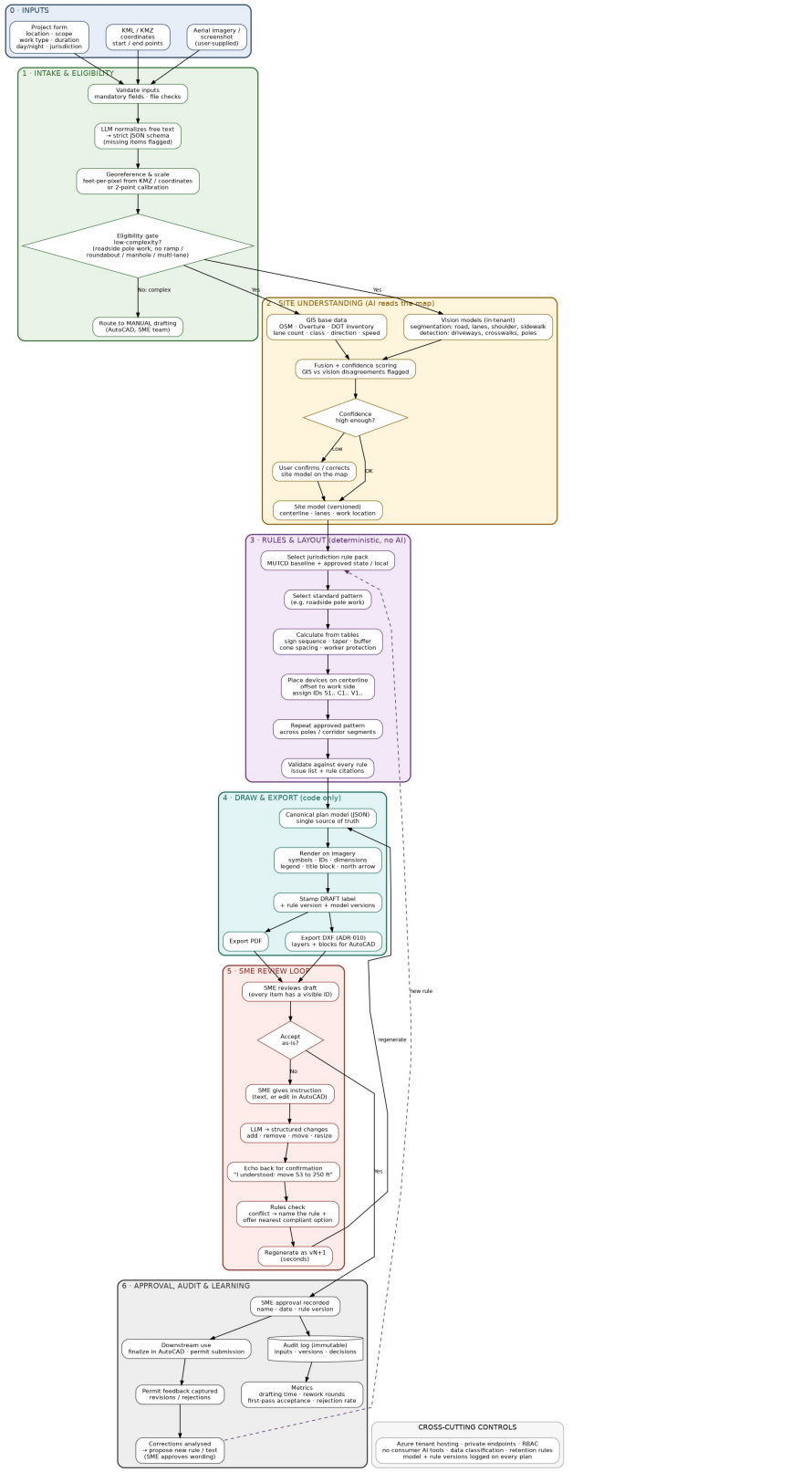

0 · INPUTS
Project form
location · scope
work type · duration
day/night · jurisdiction
KML / KMZ
coordinates
start / end points
Aerial imagery /
screenshot
(user-supplied)
1 · INTAKE & ELIGIBILITY
Validate inputs
mandatory fields · file checks
LLM normalizes free text
→ strict JSON schema
(missing items flagged)
Georeference & scale
feet-per-pixel from KMZ / coordinates
or 2-point calibration
Eligibility gate
low-complexity?
(roadside pole work, no ramp /
roundabout / manhole / multi-lane)
No: complex
Yes
Yes
Route to MANUAL drafting
(AutoCAD, SME team)
2 · SITE UNDERSTANDING (AI reads the map)
GIS base data
OSM · Overture · DOT inventory
lane count · class · direction · speed
Vision models (in-tenant)
segmentation: road, lanes, shoulder, sidewalk
detection: driveways, crosswalks, poles
Fusion + confidence scoring
GIS vs vision disagreements flagged
Confidence
high enough?
Low
OK
User confirms / corrects
site model on the map
Site model (versioned)
centerline · lanes · work location
3 · RULES & LAYOUT (deterministic, no AI)
Select jurisdiction rule pack
MUTCD baseline + approved state / local
Select standard pattern
(e.g. roadside pole work)
Calculate from tables
sign sequence · taper · buffer
cone spacing · worker protection
Place devices on centerline
offset to work side
assign IDs S1.. C1.. V1..
Repeat approved pattern
across poles / corridor segments
Validate against every rule
issue list + rule citations
4 · DRAW & EXPORT (code only)
Canonical plan model (JSON)
single source of truth
Render on imagery
symbols · IDs · dimensions
legend · title block · north arrow
Stamp DRAFT label
+ rule version + model versions
Export PDF
Export DXF (ADR-010)
layers + blocks for AutoCAD
5 · SME REVIEW LOOP
SME reviews draft
(every item has a visible ID)
Accept
as-is?
No
SME gives instruction
(text, or edit in AutoCAD)
LLM → structured changes
add · remove · move · resize
Echo back for confirmation
"I understood: move S3 to 250 ft"
Rules check
conflict → name the rule +
offer nearest compliant option
Regenerate as vN+1
(seconds)
new rule
regenerate
Yes
6 · APPROVAL, AUDIT & LEARNING
SME approval recorded
name · date · rule version
Downstream use
finalize in AutoCAD · permit submission
Audit log (immutable)
inputs · versions · decisions
Permit feedback captured
revisions / rejections
Metrics
drafting time · rework rounds
first-pass acceptance · rejection rate
Corrections analysed
→ propose new rule / test
(SME approves wording)
CROSS-CUTTING CONTROLS
Azure tenant hosting · private endpoints · RBAC
no consumer AI tools · data classification · retention rules
model + rule versions logged on every plan